

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

15 September 2014

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
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Total	

This document consists of **18** printed pages and **2** blank pages.

[Turn over]

Answer **all** the questions in this section.

- 1 Restriction enzymes are used during genetic engineering. Fig 1.1 shows restriction sites for *EcoRI*, *BamHI* and *AluI*.

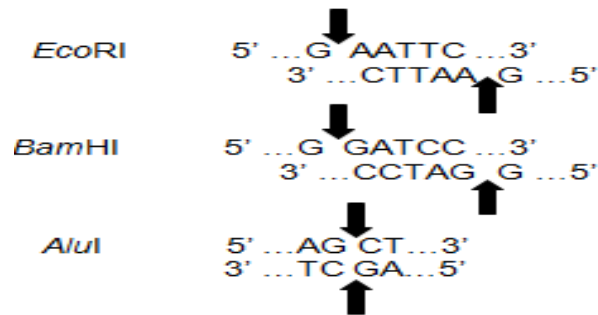


Fig. 1.1

- (a) Describe the natural function of these restriction enzymes. [3]

- (b) Explain why it is more suitable to use *EcoRI* than *AluI* in genetic engineering. [3]

A bacterial plasmid is shown in Fig. 1.2. Apart from the DNA sequence shown, there are no target sequences for the three restriction enzymes in the rest of the plasmid.



Fig 1.2

(c) With reference to Fig. 1.1 and Fig. 1.2, state the number of fragments of DNA present after the plasmid has been cut with:

(i) *EcoRI* [1]

(ii) *EcoRI* and *BamHI* [1]

Eukaryotic genes can be cloned using bacteria cells whereby the gene of interest is amplified by the bacteria as it replicates the plasmid before division. The Polymerase Chain Reaction (PCR) is another method for amplifying DNA and it is carried out *in vitro* using a thermal cycler.

The first stage of the PCR cycle involves heating the PCR mixture to cause DNA denaturation.

(d) Describe the next two stages of PCR.

(i) 2nd Stage: [2]

(ii) 3rd stage: [2]

[Total: 12]

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- 2** Severe combined immunodeficiency (SCID) represents a group of rare and sometimes, fatal disorders characterized by failure of the immune system. All forms of SCID are inherited, with X-linked SCID being the most common.

(a) Describe the genetic basis of X-linked SCID that leads to failure of the immune system. [4]

(b) The best treatment for SCID is a bone marrow transplant from an immunologically matched sibling. When no matched donor is available, unmatched donors, such as parents, are recruited; these transplants are only around 70% successful. Success of gene therapy may rival or better that seen in these unmatched donor situations.

(i) Explain what is meant by gene therapy. [2]

(ii) Explain why SCID is a suitable choice for gene therapy. [2]

- (c) In 2001, gene therapy aimed at treating X-linked SCID succeeded for 15 French baby boys. However, 3 of the little boys later developed cancer and died.

Suggest how gene therapy could have contributed to the development of cancer in the 3 boys. [2]

- (d) Another form of characterized SCID is caused by mutation of adenosine deaminase (ADA) gene.

A polymorphic DNA segment, DXS86 is found to be tightly linked to the ADA gene. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a marker for the genetic screening of ADA SCID.

Fig. 2.1 shows the restriction sites of *Bgl* II (indicated by *) on a segment of DXS86 and a probe that can be used to detect the polymorphism. Allele 2 is linked to the mutant ADA allele.

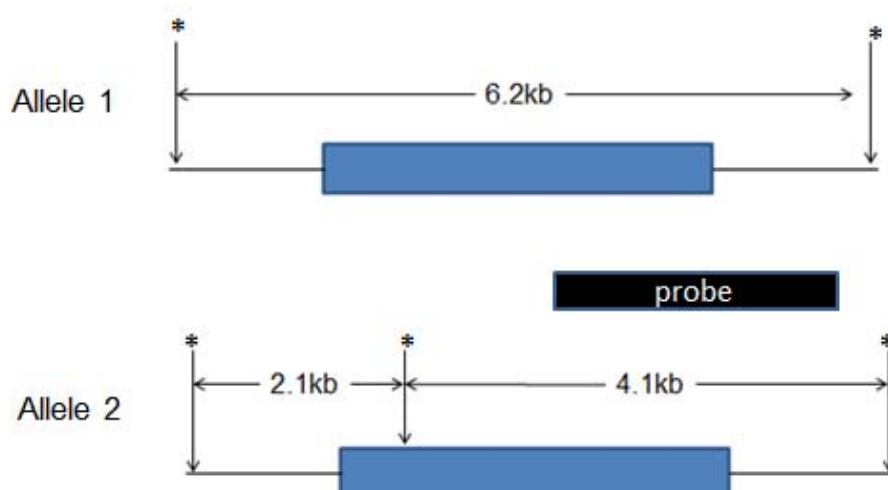


Fig.2.1

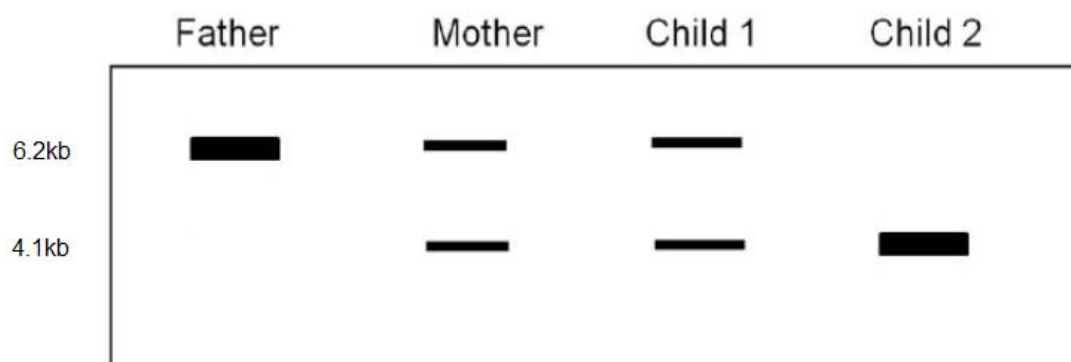


Fig. 2.2

Fig. 2.2 shows the genetic screening results for ADA SCID in a family. The amplified DNA containing DXS86 was digested with restriction enzymes followed by gel electrophoresis and Southern blotting using the probe shown in Fig. 2.1.

- (i) Based on Fig 2.1 and Fig. 2.2, explain if Child 2 is likely to get ADA SCID. [2]

- (ii) If Child 1 shows symptoms of ADA-SCID and disregarding the possibility of crossing over between DXS86 and ADA loci during gamete formation, suggest a reason for the banding pattern obtained in Child 1. [2]

[Total: 14]

- 3 Rice, *Oryza sativa*, is the staple diet in many parts of the world. It lacks a number of important nutrients, including β carotene, from which vitamin A is synthesised. Adequate concentrations of vitamin A give protection from night blindness. Higher concentrations act as an antioxidant that may give some protection from cancer and heart disease.

Golden rice, which contains β carotene, was developed by genetically modifying rice using genes from a daffodil (a flowering plant) and a bacterium.

The expression of these inserted genes is under the control of promoters found only in the endosperm (rice grain).

Fig. 3.1 shows a section of the artificial DNA sequence used.

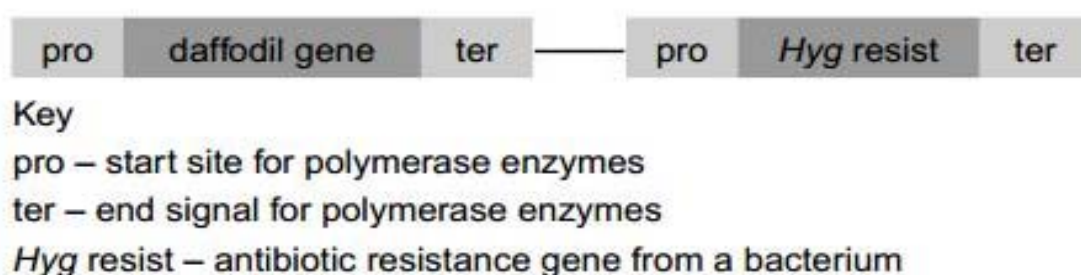


Fig. 3.1

Fig. 3.2 shows the main events in obtaining a transgenic plant.

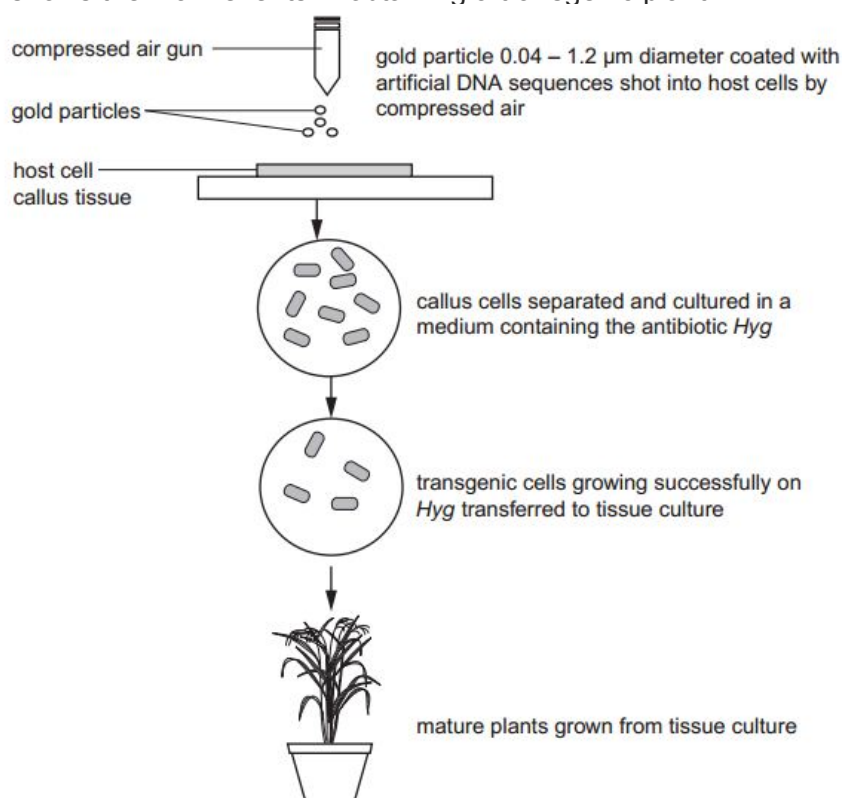


Fig. 3.2

- (a) With reference to Fig. 3.1 and Fig. 3.2,
(i) explain what is meant by 'an artificial DNA sequence'. [1]

- (ii) explain the significance of including the *Hyg* gene in the procedure. [2]

- (iii) suggest a reason for the relatively low efficiency of obtaining transgenic plants by using an air gun (gene gun) to deliver artificial DNA sequences into the plant cells. [1]

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- (b) A separate investigation was carried out into the effect of different cytokinin concentrations on the development of rice callus cultures. The results obtained after six weeks are shown in Table 3.3.

	Cytokinin concentration/ %		
	0	20	40
Mean number of roots	20	0	0
Mean number of shoots	4	15	25
Size of shoots	Small	Large	Medium

Table 3.3

- (i) Briefly explain what is meant by 'callus culture'. [1]

- (ii) With reference to data from Table 3.3, state the main conclusions that can be drawn from this investigation. [3]

Many varieties of rice, *Oryza sativa*, will die quickly if they are totally submerged by flood water (submergence-intolerance).

A plant breeder and genetics scientist has just discovered a gene, *JJ8* that was found to be involved in tolerance of prolonged submergence in rice varieties. An allele, *JJ8-A*, was subsequently found only in submergence-tolerant rice.

Varieties that are submergence-intolerant either have a second allele, *JJ8-a*, or lack the gene altogether.

Typically, naturally submergence-intolerant plants respond to submerged conditions by elongating rapidly.

(c) Suggest a reason for plants elongating rapidly when they are submerged in water. [1]

The mean heights of three types of rice plants were compared before and after 10 days of submergence in water. The types of rice plants were:

- naturally submergence-intolerant plants, denoted by **I**
- transgenic submergence-intolerant plants expressing *JJ8-A*, denoted by **I⁺**
- naturally submergence-tolerant plants, denoted by **T**.

The results are shown in Fig. 3.4.

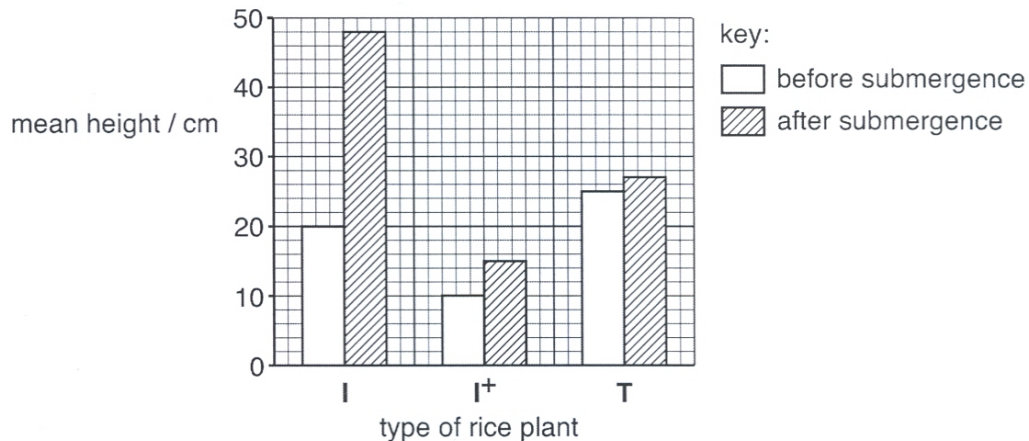


Fig.3.4

- (d) With reference to Fig. 3.4, describe the differences in mean height before and after submergence in the three types of rice plants. [3]

The allele *JJ8-A* could be introduced into a high-yielding variety of submergence-intolerant rice either by genetic engineering or by traditional selective breeding (crossing) of different rice plants.

- (e) Suggest **two** advantages of using genetic engineering, rather than traditional selective breeding, to introduce the allele. [2]

[Total: 14]

4 Planning Question

An α -amylase obtained from a species of bacterium, *Bacillus subtilis*, hydrolyses amylose to glucose.

A colorimeter is a device that can be used to measure the concentration of a solute by measuring the absorbance of different wavelengths of light in a solution. Generally, a solution of higher concentration will absorb more light energy at a specific wavelength. Therefore, the higher the absorbance value, the higher the concentration of a solution. Although absorbance has no units, it is often recorded as absorbance units (A.U.)

Design an experiment to test the hypothesis that:

The rate of glucose produced is dependent on the concentration of α -amylase.

You must use:

- 5.0% stock α -amylase solution which can be diluted to give various solutions of amylase concentration
- 1% starch solution
- Benedict's reagent
- 5 x specimen tubes
- 6 x boiling tubes
- 6 x plastic vials for containing solution to be placed in colourimeter
- 2 x 5cm³ syringe
- Bunsen burner
- Colourimeter attached to data logger
- Thermostatically controlled water bath set at 37°C

Your plan should have a clear and helpful structure to include:

- a description of the method used including scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- correct use of technical and scientific terms,
- relevant risks and precautions taken.

[Total: 12]

[illegible]

[illegible]

[illegible]

Free-response question

Write your answers to this question on the separate paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Explain how gel electrophoresis is able to separate fragments of DNA. [6]
- (b)** Explain how RFLP analysis has helped in the direct detection of sickle-cell anaemia. [6]
- (c)** Explain the benefits of the human genome project. [8]

[Total: 20]